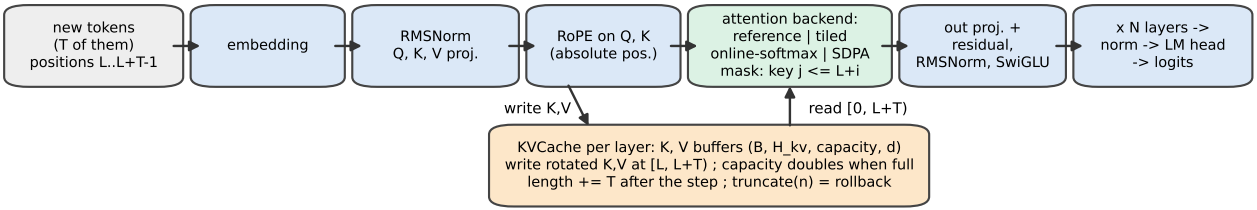
(a) One decoder step with the explicit KV cache
new tokens (T of them) positions L..L+T-1
embedding
RMSNorm Q, K, V proj.
RoPE on Q, K (absolute pos.)
attention backend: reference | tiled online-softmax | SDPA mask: key j <= L+i
out proj. + residual, RMSNorm, SwiGLU
x N layers -> norm -> LM head -> logits
write K,V
read [0, L+T)
KVCache per layer: K, V buffers (B, H_kv, capacity, d) write rotated K,V at [L, L+T) ; capacity doubles when full length += T after the step ; truncate(n) = rollback
(b) Greedy speculative decoding round (target T, draft D share the tokenizer)
Invariant: both caches hold every accepted token except 'last'
D drafts k tokens d1..dk greedily from 'last' (k cheap passes)
T verifies in ONE pass on [last, d1..dk] -> greedy g0..gk
accept longest prefix d(i+1) = g(i); emit d1..dn + g(n) (n+1 >= 1 tokens)
rollback: truncate T and D caches; if n=k feed dk to D; last <- g(n)
next round (output is identical to greedy decoding of T)